Extending QCARE: How Reward Gap Difficulty Shapes the Cost of Over and Under Exploration

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1. Introduction

The exploration-exploitation(EE) trade-off is a fundamental challenge in sequential decision making. When an agent repeatedly chooses between options with unknown reward probabilities, it has to balance gathering information about potentially better options (exploration) against maximising rewards from options that already appear superior (exploitation). The multi-armed bandit problem provides a framework for studying this trade-off.

Ding et al. (2024) propose Quantal Choice with Adaptive Reduction of Exploration (QCARE), which is a behavioural model that generalises Thompson Sampling through a single parameter α controlling how fast exploration reduces as experience accumulates. When α = 0.5, QCARE reduces to Gaussian Thompson Sampling, which is asymptotically optimal in terms of regret. Using data collected from 610 human subjects across multiple experimental rounds, Ding et al. find that humans systematically over-explore, with their implied α values clustering below 0.5, suggesting a consistent behavioural bias toward insufficient exploitation.

Ding et al. evaluate regret performance under two reward configurations: (0.4, 0.5) and (0.6, 0.4), both representing relatively small reward gaps. This report extends their simulation analysis across a wider range of reward gap difficulties, examining how the regret cost of over and under exploration varies with problem difficulty. I further identify the empirically optimal α for each configuration under a finite time horizon of T=200 rounds.

2. The QCARE Model

In the two-armed bandit problem, an agent chooses between two arms at each round t, observing a binary reward of 1 or 0. The true reward probability of each arm is unknown and is learnt through repeated pulls.

QCARE assigns a score to each arm at every round:

$$\theta_i(t) = \hat{\mu}_i(t) + \frac{1}{(k_i(t)+1)^{\alpha}}\epsilon_{it}$$

where:

- $\theta_i(t)$ = score assigned to arm i at round t
- $\hat{\mu}_i(t)$ = empirical mean reward of arm i
- $k_i(t)$ = number of times arm i has been pulled
- $\varepsilon_i t \sim N(0,1)$ = standard normal random shock
- α = reduction rate of exploration

The empirical mean captures exploitation, favouring arms with better historical performance. The noise term captures exploration, occasionally favouring arms that have not been pulled much. As $k_i(t)$ grows, the noise shrinks, naturally reducing exploration over time.

A larger α causes the noise term to shrink faster with each pull, leading the agent to commit to the leading arm more quickly. A smaller α keeps the noise large for longer, sustaining exploration even as evidence accumulates.

When α = 0.5, QCARE is equivalent to Gaussian Thompson Sampling. Any value below 0.5 represents over-exploration while any value above 0.5 represents under-exploration relative to Thompson Sampling.

3. Simulation Setup

This study extends the simulation analysis of Ding et al. across a wider range of reward configurations. Rather than estimating α from human behavioural data, α is fixed across a range(pre-defined) to examine how regret varies systematically with exploration tendency. While of course real human behaviour cannot be reduced to a single fixed α, this approach allows for controlled analysis of how the cost of over and under exploration shifts across different reward environments.

Reward configurations tested:

I examined four reward gap configurations:

- (0.45, 0.55) → smallest gap(0.1)
- (0.40, 0.60) → small gap(0.2)
- (0.30, 0.70) → moderate gap(0.4)
- (0.10, 0.90) → largest gap(0.8)

In each configuration arm 2 has the higher reward probability. The gap between arms ranges from 0.1 to 0.8, extending well beyond the configurations tested by Ding et al.

Parameter settings:

- α looped from 0.1 to 2.0 in increments of 0.1
- T = 200 rounds per simulation
- 5,000 simulations per condition
- Regret measured as cumulative expected reward loss from not always pulling the optimal arm

4. Results

Figure 1 shows the average regret as a function of α across the four reward gap configurations. Two patterns emerge clearly.

First, for all configurations, regret is minimised near α = 0.5 to 1.0, consistent with Ding et al.'s theoretical result that Thompson Sampling (α = 0.5) is asymptotically optimal. However, the optimal α in these finite horizon simulations (T = 200) sits above 0.5 for all configurations, suggesting that agents should exploit more aggressively than Thompson Sampling prescribes in short horizon settings.

Second, the cost of over-exploration (α < 0.5) increases sharply with reward gap size. For the largest gap configuration (0.1, 0.9), regret at α = 0.1 reaches approximately 34, compared to roughly 9 for the smallest gap (0.45, 0.55). This occurs because each pull of the suboptimal arm is more costly when the gap between arms is large.

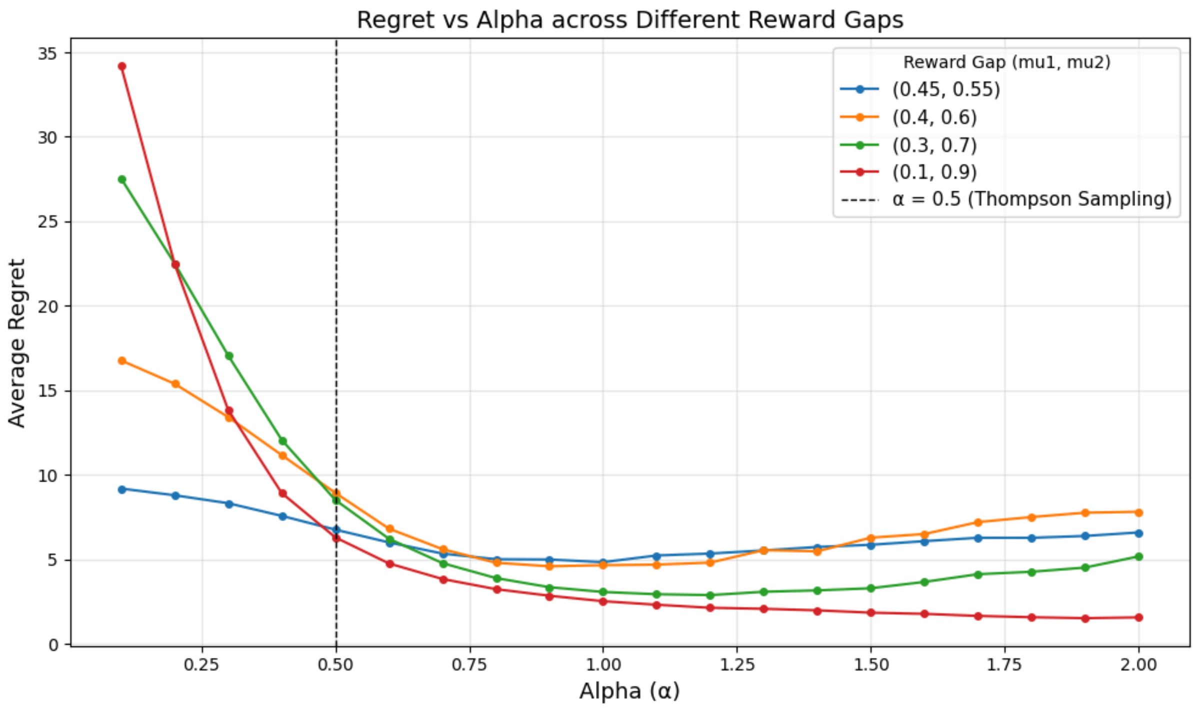

Figure 1: Regret vs Alpha across Different Reward Gaps

Figure 2 shows the empirically optimal α for each reward gap configuration. The optimal α increases with gap size, rising from approximately 1.0 at gap 0.1 to nearly 1.9 at gap 0.8, suggesting that larger reward gaps warrant more aggressive exploitation.

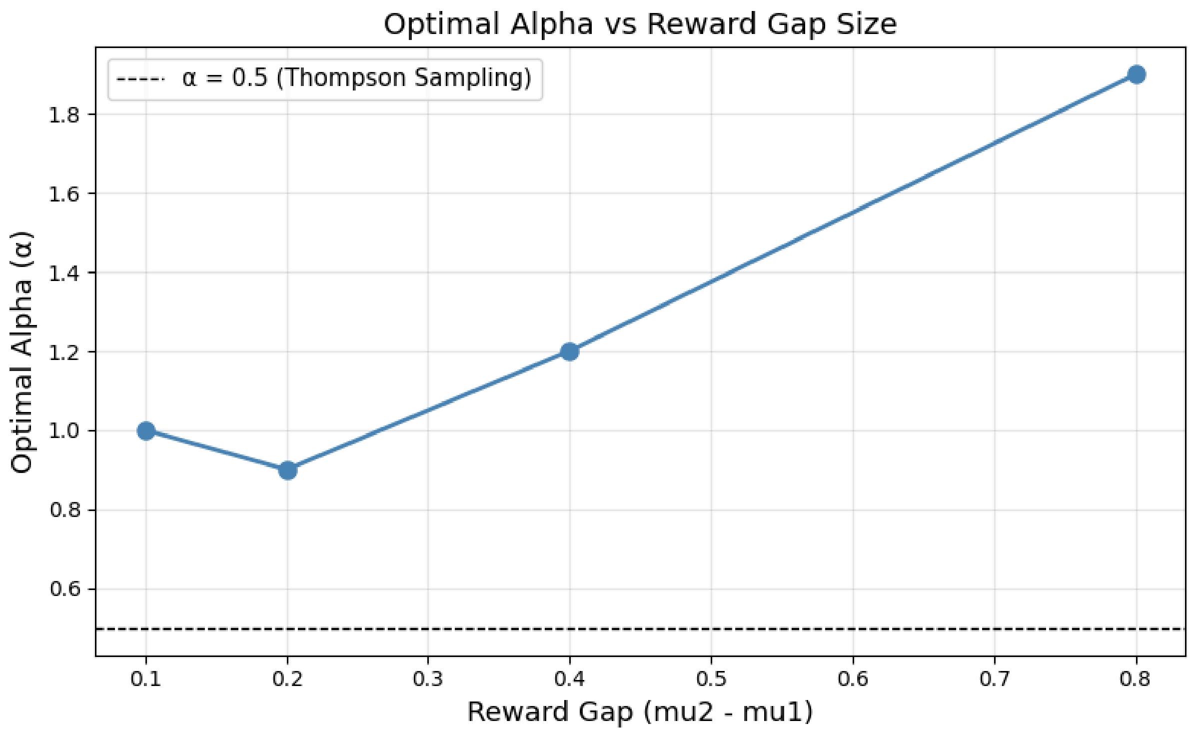

Figure 2: Optimal Alpha vs Reward Gap Size

Taken together, these findings suggest that the optimal exploration strategy is not universal but depends on the difficulty of the environment.

5. Discussion and Future Work

The results reveal two key insights that extend the findings of Ding et al.

First, the cost of over-exploration is not uniform across environments. Ding et al. find that humans systematically over-explore with implied α values below 0.5. However, the simulations suggest that this behavioural tendency is far more damaging in easy environments, where the reward gap is large and the correct arm is relatively obvious compared to hard environments where the arms do not have much between them. In easy environments, each wrong pull is expensive and the agent should be exploiting a lot more. In hard environments, continued exploration is more forgivable because the cost per wrong pull is small.

Second, the optimal α shifts with reward gap difficulty. In finite horizon settings, Thompson Sampling (α = 0.5) is not universally optimal. Agents in easy environments should exploit more aggressively than Thompson Sampling prescribes. This is consistent with findings in Ding et al. who note that Thompson Sampling tends to over-explore in finite horizons.

These findings raise an important question about Ding et al.'s experimental design. Their human subjects played bandit problems with reward gaps of 0.1 and 0.2 which are relatively hard problems. The over-exploration they observe may be less concerning in those specific environments than it would be in easier real-world decision settings with more obvious reward differences.

Future Work

Several extensions of this analysis are worth pursuing. First, the current simulation assumes binary Bernoulli rewards. During the seminar at the Singapore University of Technology and Design(SUTD) of this paper, an audience member raised the question of whether QCARE's properties extend to non-Bernoulli reward distributions(Gaussian, heavy-tailed, etc.). The speaker (Prof Yifan Feng) noted this as a technical challenge and that extending the maximum likelihood estimation (MLE) and theoretical guarantees to continuous reward distributions remains an open problem.

Second, this study treats α as a fixed parameter. A natural extension would examine whether α adapts dynamically within a single decision maker. For instance, whether the same person explores more patiently in hard environments and exploits more aggressively in easy ones. If α is context-dependent rather than a fixed trait, the behavioural implications of Ding et al.'s findings would need to be reinterpreted.

References

Ding, J., Feng, Y., & Rong, Y. (2024). A behavioral model for exploration vs. exploitation: Theoretical framework and experimental evidence. *Management Science* (forthcoming). https://arxiv.org/pdf/2207.01028